Sharp Norms for Involutory Dilations

Candidate full answers to Questions 3.2 and 3.4 of arXiv:2602.19599

Autonomous proof candidate for human review

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Status. Candidate full resolution, likely valid. The involution already displayed in Proposition 3.3 of the source has the exact operator norm $\|A\| + \sqrt{1 + \|A\|^2}$. This gives a real-preserving dilation in dimension $2n$, improves the constant 3 sharply to $1 + \sqrt{2}$, and, apart from the elementary real one-dimensional exception described below, attains the smallest possible universal norm.

1 The questions and the answer

For a matrix $A \in M_n(\mathbb{F})$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, the source introduces the involution

$$S_A = \begin{pmatrix} A & I + A \\ I - A & -A \end{pmatrix} \in M_{2n}(\mathbb{F}) \tag{1}$$

and proves $\|S_A\| \leq 1 + 2\|A\|$. Question 3.2 asks whether a real contraction has a real involutory dilation with the sharp norm $1 + \sqrt{2}$. Question 3.4 asks for the sharp replacement of 3 in the bound for S_A, and whether that replacement is also the optimal norm for arbitrary involutory dilations in dimension $2n$.

Theorem 1. *For every $A \in M_n(\mathbb{F})$,*

$$\boxed{\|S_A\| = \|A\| + \sqrt{1 + \|A\|^2}}. \tag{2}$$

Consequently:

1. *every contraction has an involutory dilation in $M_{2n}(\mathbb{F})$ of norm at most $1 + \sqrt{2}$, with no change of scalar field;*
2. *the sharp constant replacing 3 for the family S_A is $\omega = 1 + \sqrt{2}$;*
3. *the least universal norm for involutory dilations in $M_{2n}(\mathbb{C})$ is $1 + \sqrt{2}$. The same is true in $M_{2n}(\mathbb{R})$ when $n \geq 2$.*

For real $n = 1$ only, the least universal norm in the last assertion is 1, although the sharp uniform bound for the specific matrices S_A remains $1 + \sqrt{2}$.

Thus Question 3.2 is answered affirmatively in a stronger dimension $2n$ form. If the statement is read as requiring exactly dimension $4n$ and norm exactly $1 + \sqrt{2}$, both can also be retained; see Corollary 1.

This constant cannot be diminished as i would not be any longer in $W(S)$, and since $W(S) \supset W(A)$, a contraction A with i in its spectrum could not be dilated into S. $\square$

In this proof, it seems necessary to use a complex matrix S even if we start with a real matrix A. Hence, we ask the following.

Question 3.2. In the previous proposition, is it possible to take S with only real entries when so is A ?

and, if A is a contraction, this is maximized with $6n$. $\square$

Question 3.4. The constant $\sqrt{6n}$ in Proposition 3.3 is the best possible one. However, it seems possible to slighly diminish the constant 3. What is the sharp value ω ? Is then ω the smallest possible operator norm ensuring that any contraction in $\mathbb{M}_n$ can be dilated into some matrix in $\mathbb{M}_{2n}$ with norm less or equal than ω ?

We may use in Proposition 3.3 the partial transpose

Figure 1: Questions 3.2 and 3.4 in the source, PDF pages 11 and 12.

2 A two-line unitary reduction

Set

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ I & -I \end{pmatrix}.$$

This matrix is orthogonal (and unitary), self-adjoint, and a direct block multiplication gives

$$HS_AH = Q_A := \begin{pmatrix} I & 0 \\ 2A & -I \end{pmatrix}. \tag{3}$$

In particular, $S_A^2 = I$ is also transparent from this form.

Lemma 1. *For every operator A on a finite-dimensional real or complex Hilbert space, if $a = \|A\|$, then*

$$\left\| \begin{pmatrix} I & 0 \\ 2A & -I \end{pmatrix} \right\| = a + \sqrt{1+a^2}.$$

Proof. For vectors x, y, put $p = \|x\|$ and $q = \|y\|$. The triangle inequality gives

$$\begin{aligned} \|Q_A(x,y)\|^2 &= \|x\|^2 + \|2Ax - y\|^2 \\ &\le p^2 + (2ap+q)^2 = \begin{pmatrix} p & q \end{pmatrix} \begin{pmatrix} 1+4a^2 & 2a \\ 2a & 1 \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix}. \end{aligned}$$

The largest eigenvalue of the displayed scalar matrix is

$$1 + 2a^2 + 2a\sqrt{1+a^2} = \big(a + \sqrt{1+a^2}\big)^2.$$

This proves the upper bound. If $a > 0$, choose a unit vector x with $\|Ax\| = a$, put $z = Ax/a$, and restrict to pairs $(rx, -tz)$ with $r, t \ge 0$. On this two-dimensional set the preceding quadratic form is attained exactly, so its maximal Rayleigh quotient gives the matching lower bound. If $a = 0$, then $Q_A = \operatorname{diag}(I, -I)$, and the formula again holds. $\square$

Because H is unitary, equations (3) and Lemma 1 prove (2). Since the upper-left corner of S_A is A, it is an involutory dilation of A. It has the same real or complex entries as A, proving the upper-bound and real-preservation assertions. The function $a \mapsto a + \sqrt{1+a^2}$ is increasing, and equality at $a = 1$ shows that the uniform constant for S_A is sharply $1 + \sqrt{2}$.

3 Optimality among all involutory dilations

We include the short lower-bound argument to distinguish the norm of the specific formula S_A from the optimum over all possible dilations. If $A = V^*SV$ is a compression, then $W(A) \subseteq W(S)$. For an involution of norm s, Corollary 2.3(d) of the source gives the numerical range as an ellipse whose vertical semiaxis is

$$\frac{s - s^{-1}}{2}. \tag{4}$$

Over $\mathbb{C}$, take the contraction $A = iI_n$. Then $i \in W(A)$, so (4) is at least 1. Hence $s - s^{-1} \geq 2$, equivalently $s \geq 1 + \sqrt{2}$. Formula (1) supplies the matching $2n$-dimensional upper bound.

For real $n \geq 2$, take

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \oplus 0_{n-2}.$$

Complexifying any proposed real involutory dilation preserves its norm and compression relation, while the complexification of A has eigenvalue i. The same lower bound therefore applies. When $n = 1$, however, each scalar contraction $a \in [-1, 1]$ has the orthogonal involutory dilation

$$R_a = \begin{pmatrix} a & \sqrt{1-a^2} \\ \sqrt{1-a^2} & -a \end{pmatrix}, \qquad R_a^2 = I, \quad \|R_a\| = 1.$$

No involution can have norm below its spectral radius 1, proving the stated exceptional value.

Corollary 1. *If $A \in M_n(\mathbb{R})$ is a contraction, there is a real involution $\widetilde{S} \in M_{4n}(\mathbb{R})$ dilating A with $\|\widetilde{S}\| = 1 + \sqrt{2}$.*

Proof. Let

$$T = \begin{pmatrix} 0 & 1+\sqrt{2} \\ \sqrt{2}-1 & 0 \end{pmatrix}.$$

Then $T^2 = I$ and $\|T\| = 1 + \sqrt{2}$. The real block direct sum $\widetilde{S} = S_A \oplus T^{\oplus n}$ has size $4n$, still contains A as its leading compression, and has the required norm. $\square$

4 Verification, novelty, and scope

The accompanying script `code/verify_involutory_dilation.py` checks $S_A^2 = I$ and the Hadamard identity exactly over random rational matrices, then compares the singular-value norm with (2) for real and complex matrices over several sizes and scales. It also verifies the exact scalar characteristic polynomial used in Lemma 1.

A bounded search on 11 August 2026 found only version 1 of the source and no later arXiv paper stating (2) or answering Questions 3.2 and 3.4. The construction (1), the numerical-range ellipse, and the optimal lower-bound example are all already present in the source; the new step claimed here is the fixed Hadamard reduction and exact norm calculation, together with its consequences. Priority is not asserted.